

A Scientific Machine Learning Study of Particle Dynamics in the Generalised Ellis–Bronnikov Wormhole

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Abstract

We study neural approximations to the motion of a particle constrained to a rotating Archimedean spiral in a Generalised Ellis–Bronnikov wormhole. Although the reduced dynamics have only two state variables, the learning problem is not uniform: the timelike velocity interval changes with position, low-energy trajectories remain close on the incoming branch before separating near the throat, and small local errors can accumulate under recursive prediction. We first map the position-dependent admissible interval to a fixed coordinate $\xi \in (-1, 1)$ and examine neural models of the local flow. Deeper local networks are more reliable than the initial shallow model, and supplying the conserved energy $\mathcal{E}_0$ as a fixed trajectory label reduces the mean absolute recursive ξ error of nine difficult incoming rollouts by 94.5%, while removing the remaining physical exit. Two losses designed to reduce local energy-normal and sensitivity-weighted errors improve the quantities they directly target, but do not improve the complete recursive trajectories. We therefore construct a direct finite-time model of the state after an arbitrary elapsed time. Writing the targets as $\Delta x = sV_x$ and $\Delta \xi = g(s)F_\xi$, with $g(s) = -5\exp(-s/5)$, enforces the identity map exactly and gives a better balance between short- and long-time errors than either raw accumulated increments or a pure average-rate representation. On a sealed set of 1,024 held-out orbits, the dense direct-prediction RMSE is 0.06295 in x and 0.002932 in ξ , with no timelike-admissibility violations. The local model remains more accurate at its native interval $s = 0.2$, but its recursively accumulated x error is overtaken by direct prediction at long times. Derivative supervision reduces error in rapid-change regions, although the remaining low-energy error is mostly tangent to the exact phase-space trajectory. A parameter-controlled late split into separate output heads does not improve that residual across the validation population. The results separate local accuracy, recursive stability, and finite-time prediction as related but distinct requirements for learned physical dynamics.

1. Introduction

The physical system considered here is the constrained particle dynamics studied by Giorgadze and Osmanov [1]. The background is the static, spherically symmetric Generalised Ellis–Bronnikov wormhole,

$$ds^2 = -dt^2 + dl^2 + R(l)^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad R(l) = (b_0^m + l^m)^{1/m},$$

where l is the proper radial coordinate, b_0 is the throat radius, and m is an even integer that controls the geometry of the wormhole function. The particle is constrained to a rotating conical Archimedean spiral,

$$\theta = \text{const}, \quad \phi(t, l) = \alpha l + \omega t.$$

This constraint is the main simplification for the present study. Once the polar and azimuthal motion have been prescribed by the spiral, l is the only remaining independent spatial coordinate. The four-dimensional problem then reduces to a stationary effective metric in (t, l) , and the motion can be written as a radial first-order system for the state (l, u) , where

$$u = \frac{dl}{dt}.$$

The corresponding phase space is restricted by relativistic admissibility. In units with $c = 1$, the total speed measured with respect to laboratory time is

$$v_{\text{tot}}^2 = u^2 + R(l)^2 \sin^2 \theta (\omega + \alpha u)^2,$$

and a massive trajectory must remain in the timelike region $v_{\text{tot}}^2 < 1$. Thus the allowed radial velocity is not fixed independently of position: its upper and lower limits vary with l . Since the reduced metric is stationary, the motion also has a conserved specific energy,

$$\mathcal{E} = -\gamma (g_{00} + g_{01}u), \quad \gamma = (-g_{00} - 2g_{01}u - g_{11}u^2)^{-1/2}.$$

The closed radial equation, the timelike condition, and the energy integral together give a completely specified low-dimensional dynamical system. Starting from any admissible initial state, one can integrate the equations with high

numerical accuracy and, in practice, generate as many reference trajectories as are required. This makes the problem useful for studying learned dynamics: the reference vector field, the physically allowed phase-space region, a conserved quantity, and accurate numerical solutions are all known independently of the learned model.

Figure 1 shows a representative family of exact trajectories in the original (l, u) phase space. The admissible interval changes with position, while trajectories belonging to different energy levels can follow markedly different paths through the same physical region. The particular parameter values and the choice of trajectory family are specified below.

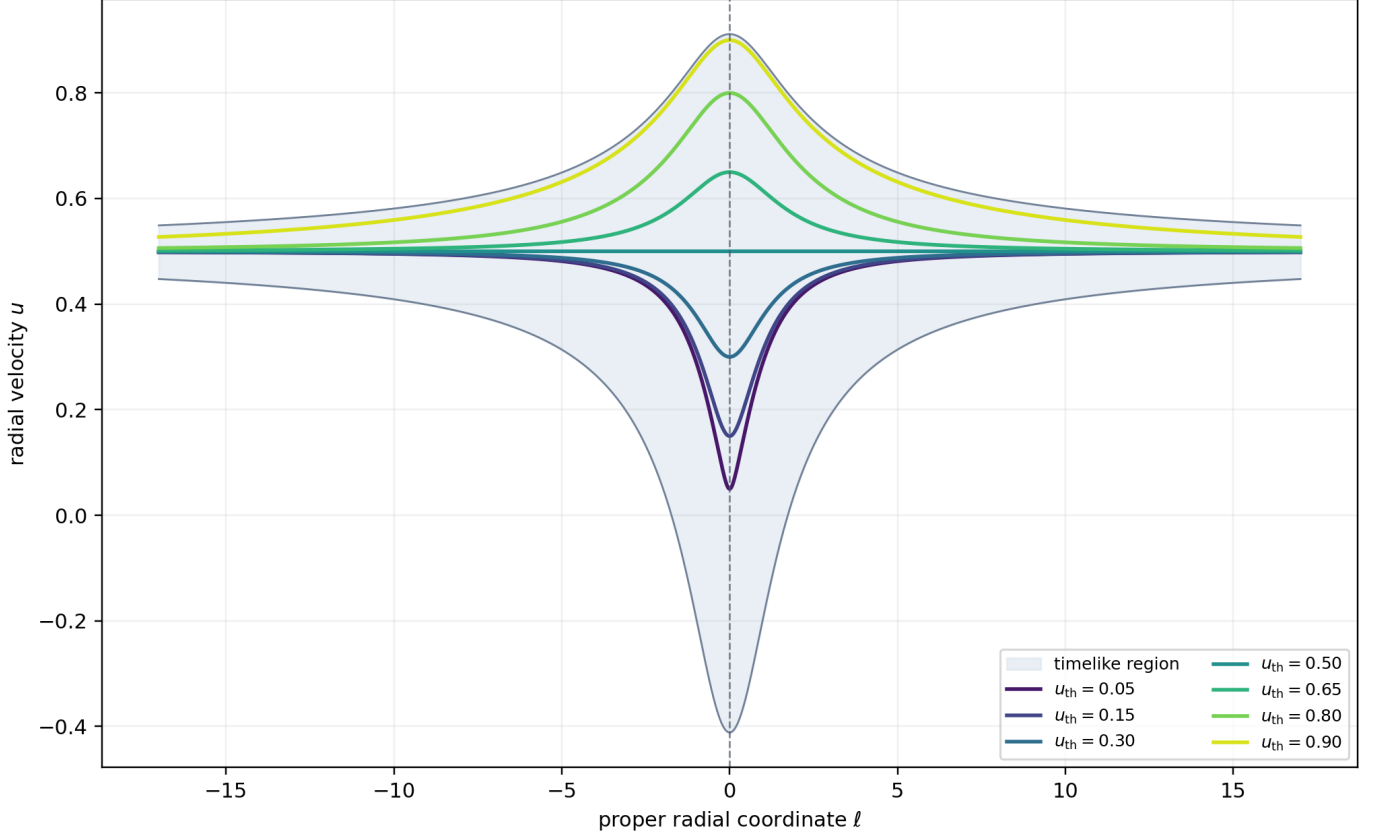


Figure 1: Exact trajectories in the original (l, u) phase space for the fixed configuration used in this study. The shaded region is timelike, its boundary is null, and the dashed vertical line marks the wormhole throat. Each curve is labelled by its radial velocity at the throat.

2. The learning problem

The numerical experiments are restricted to one fixed wormhole and spiral configuration,

$$b_0 = 1, \quad m = 2, \quad \omega = 1, \quad \alpha = -2, \quad \theta = \frac{\pi}{6}.$$

We use the dimensionless variables

$$x = \frac{l}{b_0}, \quad s = \frac{t}{b_0}, \quad u = \frac{dl}{dt}.$$

Rescaling l and t by the same length scale leaves the numerical value of u unchanged in units with $c = 1$. Moreover, because $b_0 = 1$ in the chosen configuration, the numerical values of (x, s) coincide with those of (l, t) . We nevertheless retain the dimensionless notation throughout, rather than identifying the variables themselves. The working spatial interval is

$$-17 \leq x \leq 17,$$

and the initial local prediction problem uses the fixed dimensionless time interval $h = 0.2$.

The numerical values used throughout the report are collected in Table 1. The tighter tolerances refer to the complete-orbit banks used for the finite-time study; the initial local reference calculations used relative and absolute tolerances 10^{-10} and 10^{-12} , respectively.

Table 1. Fixed physical and numerical configuration.

Quantity	Value
Throat radius b_0	1
Wormhole parameter m	2
Spiral angular velocity ω	1
Spiral pitch α	-2
Cone angle θ	$\pi/6$
Spatial interval	$-17 \leq x \leq 17$
Local prediction interval h	0.2
Complete-orbit solver	DOP853
Complete-orbit tolerances	relative 10^{-11} , absolute 10^{-13} , maximum step 0.2
Retained throat-velocity range	$0.01 \leq u_{\text{th}} \leq 0.9048234494884861$

The radial velocity cannot be sampled from one position-independent interval. Writing $R(x) = (1 + x^m)^{1/m}$ for the areal radius in units of b_0 , and defining

$$q(x) = R(x)^2 \sin^2 \theta,$$

the timelike condition may be written as the quadratic inequality

$$(1 + q\alpha^2) u^2 + 2q\omega\alpha u + q\omega^2 - 1 < 0.$$

Its two roots can be expressed as

$$u_-(x) = c(x) - d(x), \quad u_+(x) = c(x) + d(x),$$

where

$$c(x) = -\frac{q\omega\alpha}{1 + q\alpha^2}, \quad d(x) = \frac{\sqrt{1 + q(\alpha^2 - \omega^2)}}{1 + q\alpha^2}.$$

The admissible velocity at a given position therefore satisfies $u_-(x) < u < u_+(x)$. We incorporate this changing corridor directly into the phase-space coordinates by defining

$$\xi = \frac{u - c(x)}{d(x)}, \quad u = c(x) + d(x)\xi.$$

The full timelike region is then mapped to the fixed strip

$$-1 < \xi < 1.$$

This transformation is determined by the physical null boundaries and has an exact inverse. It is not an ordinary statistical normalization fitted to a collection of samples. In particular, the same value of ξ specifies the same relative position within the local admissible corridor, even though the corresponding physical velocity changes with x . The coordinate therefore removes the changing width and centre of the physical domain from the representation while preserving the original radial velocity.

For the trajectory family used later, the particle enters from $x = -17$ with positive radial velocity, crosses the throat once, and leaves into $x > 0$, with the integration continued until $x = 17$. A convenient physical label for these trajectories is the radial velocity at the throat,

$$u_{\text{th}} = u(x = 0) > 0.$$

At the fixed parameter values above, u_{th} labels the conserved-energy branch and hence distinguishes trajectories that may be very close in part of the (x, ξ) plane. The retained range is

$$0.01 \leq u_{\text{th}} \leq 0.9048234494884861.$$

Figure 2 shows seven reference trajectories drawn from this family in the transformed phase space. Several of the lower- u_{th} trajectories are compressed into a narrow region on the incoming branch, where their values of both u and ξ differ only slightly. As the particle approaches the throat, the same trajectories separate much more strongly. Thus proximity in the incoming local phase space does not imply similarly close behaviour over the remainder of the traversal. This geometry is already a property of the exact dynamics; no learned model is involved in the comparison.

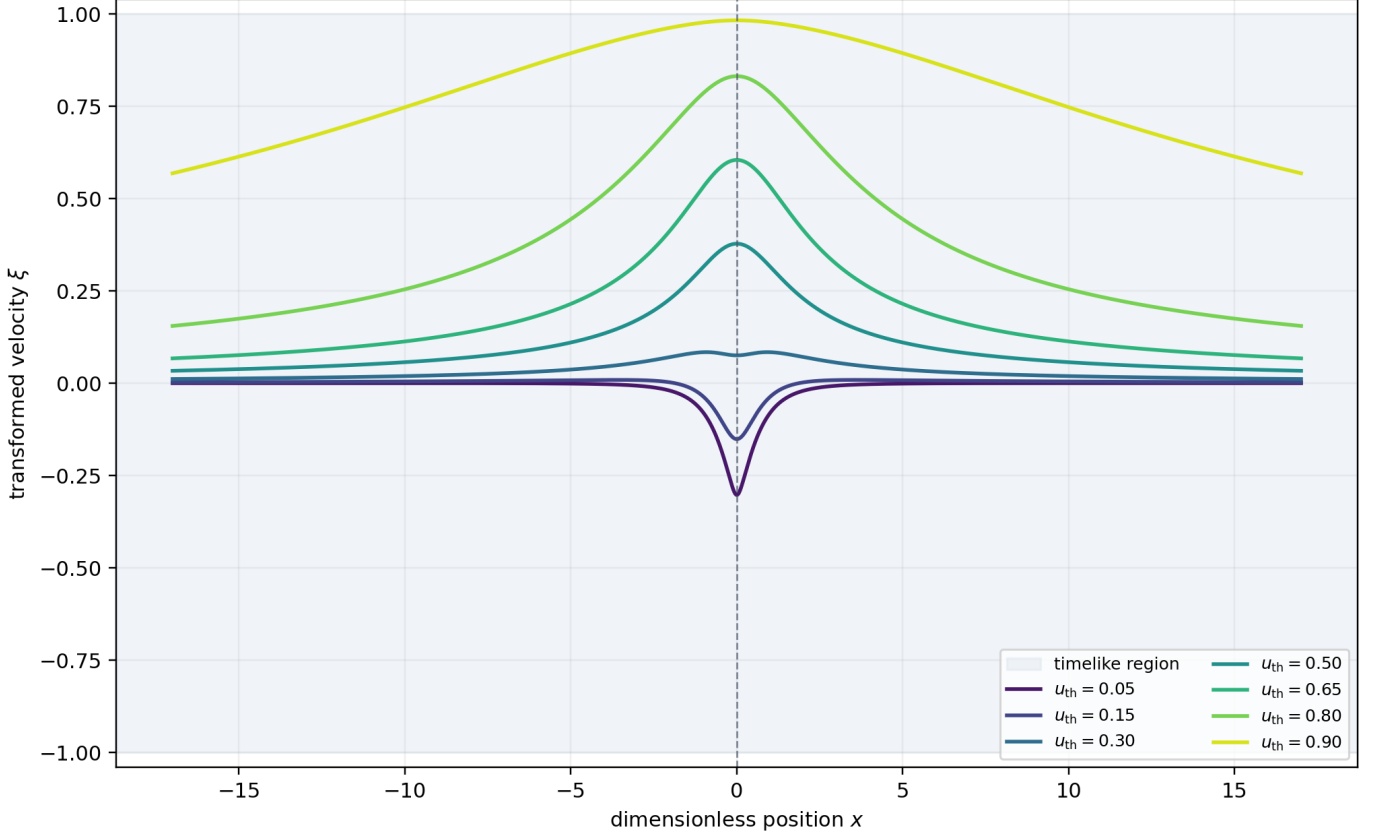


Figure 2: Exact trajectories in the transformed (x, ξ) phase space, labelled by their throat velocity u_{th} . The transformation maps the complete timelike region to $-1 < \xi < 1$. The lower- u_{th} trajectories remain close over much of the incoming branch before separating near the throat.

3. Local predictions, rollouts and conserved energy

The first learning task was to approximate the exact evolution over a fixed interval h . Let z_n denote the chosen representation of the state at time s_n . The exact flow map is

$$\Phi_h : z_n \mapsto z_{n+1}, \quad z_{n+1} = \Phi_h(z_n),$$

and the corresponding increment is

$$\Delta z_n = \Phi_h(z_n) - z_n.$$

The learned map predicts this increment and is written in residual form. When it is used recursively, the state evolves according to

$$\hat{z}_{n+1} = \hat{z}_n + \Delta z_\vartheta(\hat{z}_n).$$

The first prediction is made from an exact initial state, whereas every later input is supplied by the model itself. A one-step validation error therefore measures the approximation of the flow from known physical states. A recursive trajectory also depends on where the preceding predictions have placed the particle, and small differences in the local map can consequently acquire a much larger effect after repeated application.

All local maps considered here were represented by fully connected feed-forward networks. Let q denote the input supplied to a particular model: for example $q = (x, u)$, (x, u, ξ) , or $(x, \xi, \mathcal{E}_0)$. Each input component and each target increment were standardized using means and standard deviations calculated from the training set,

$$\tilde{q}_i = \frac{q_i - \mu_{q_i}}{\sigma_{q_i}}, \quad \widetilde{\Delta z}_j = \frac{\Delta z_j - \mu_{\Delta z_j}}{\sigma_{\Delta z_j}}.$$

For a network with L hidden layers, the calculation through hidden layer r is

$$a^{(r)} = W^{(r)}h^{(r-1)} + b^{(r)}, \quad h^{(r)} = \tanh(a^{(r)}), \quad h^{(0)} = \tilde{q},$$

If hidden layer r contains n_r units, then

$$W^{(r)} \in \mathbb{R}^{n_r \times n_{r-1}}, \quad b^{(r)} \in \mathbb{R}^{n_r}.$$

The output layer is linear,

$$\widetilde{\Delta z} = W^{(L+1)}h^{(L)} + b^{(L+1)},$$

and the physical increment used in the residual update is recovered as

$$\Delta z_\vartheta(q) = \mu_{\Delta z} + \sigma_{\Delta z} \odot \widetilde{\Delta z},$$

with $\odot$ denoting componentwise multiplication. The weights $W^{(r)}$ and biases $b^{(r)}$ are the trainable parameters. If the successive layer widths are $n_0, n_1, \dots, n_{L+1}$, their total number is

$$N_{\text{par}} = \sum_{r=1}^{L+1} (n_r n_{r-1} + n_r).$$

Training minimizes the mean squared error of the standardized increments,

$$\mathcal{L}(\vartheta) = \frac{1}{N} \sum_{k=1}^N \left\| \widetilde{\Delta z}_\vartheta(\tilde{q}_k) - \widetilde{\Delta z}_k \right\|_2^2.$$

This definition is common to the architecture and input comparisons below. What changes between the models is the choice of q and the number and width of the hidden layers; the predicted dynamical quantity remains the local state increment.

The initial architecture comparison used the same 20,000 training states and 4,000 validation states, with every target obtained by evolving an initial state through $h = 0.2$. Inputs and target increments were standardized using the training set only. Each model was trained for seeds 101, 202, and 303 with Adam, a learning rate of 10^{-3} , batches of 512 states, and a maximum of 1,000 epochs with early-stopping patience 40.

Three models summarize the useful part of this early progression. The first was a shallow physical-coordinate model,

$$(x, u) \mapsto (\Delta x, \Delta u), \quad 2 \longrightarrow 64 \longrightarrow 2,$$

with 322 trainable parameters. We denote it by A64. The second retained the same inputs and targets but used two hidden layers,

$$(x, u) \mapsto (\Delta x, \Delta u), \quad 2 \longrightarrow 32 \longrightarrow 32 \longrightarrow 2,$$

giving 1,218 parameters; this is A32 $\times$ 32. The third supplied the bounded coordinate ξ in addition to (x, u) ,

$$(x, u, \xi) \mapsto (\Delta x, \Delta u), \quad 3 \longrightarrow 32 \longrightarrow 32 \longrightarrow 2,$$

with 1,250 parameters; this is C32 $\times$ 32. The hidden activations were tanh in each case.

The recursive comparison is summarized in Table 2. Its error is the combined state error after scaling x and u by their training-set scales, averaged over the stored times and validation trajectories and then across the three seeds. A physical exit means that a predicted state crossed the timelike boundary. It is a failure of the unconstrained learned trajectory, not of the numerical reference solution.

Table 2. Recursive-rollout results for the three representative early architectures.

Model	Inputs	Hidden layers	Parameters	Physical exits / 72	Mean stored rollout error
A64	(x, u)	64	322	25	0.04324
A32×32	(x, u)	32, 32	1,218	0	0.02354
C32×32	(x, u, ξ)	32, 32	1,250	0	0.02163

The shallow A64 model left the physical region in 25 of the 72 rollout cases. Adding depth reduced the mean trajectory error and removed these exits for A32×32. Supplying ξ to the comparably sized C32×32 model gave a smaller further reduction in the mean error while retaining zero exits. Figure 3 shows the same progression for a representative trajectory that crosses the throat. The numerical solution remains inside the admissible phase space. One of the shallow-model predictions instead crosses its boundary, while the other predicted curves differ in the velocity reached during the central passage. The deeper models remain physical in this example, although their trajectories are not identical to the reference and still depend visibly on the training seed.

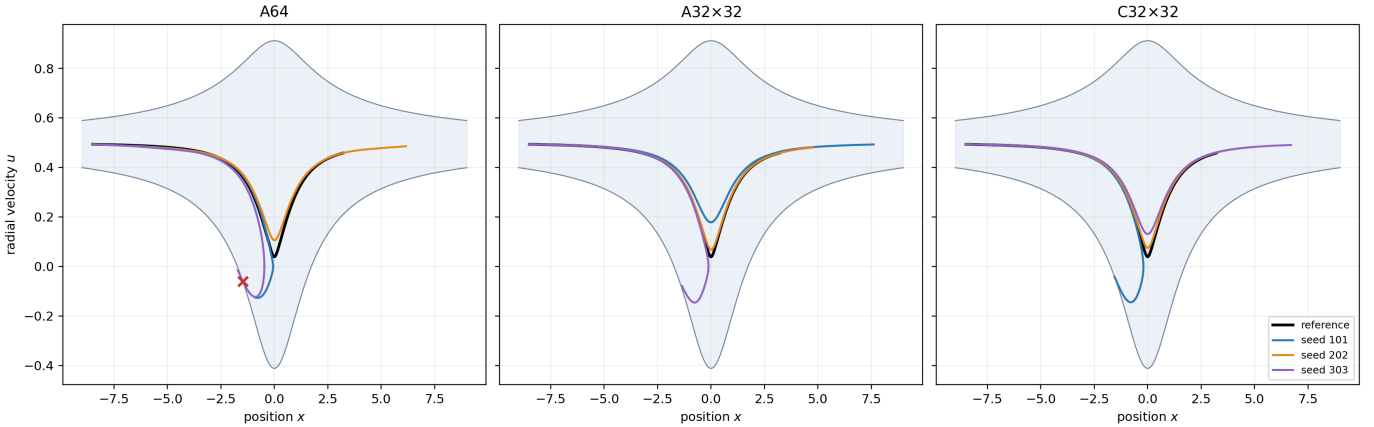


Figure 3: Exact and recursively predicted trajectories for a representative throat-crossing case. The black curve is the DOP853 reference, and the shaded region is timelike. One A64 prediction crosses the admissible boundary, whereas the two deeper models remain physical but retain seed-dependent differences through the central region.

This progression should not be read as a capacity-only comparison. A64 and A32×32 differ in depth as well as parameter count, while C32×32 also changes the input representation. The conclusion is more limited: increasing the width of a shallow network was not sufficient for reliable recursive prediction, deeper structure helped substantially, and the additional physical coordinate produced a further but more modest improvement. Network size alone did not determine the quality of the learned trajectory, and one-step performance did not by itself describe the behaviour obtained after many recursive steps.

The later local study retained the two-hidden-layer structure but expressed both the state and the predicted increment in the transformed coordinates,

$$(x, \xi) \mapsto (\Delta x, \Delta \xi).$$

The corresponding model used 40,000 training states and 8,000 validation states, with additional resolution of the narrow incoming region. This produced small one-step errors, but the difficult low- u_{th} trajectories still exposed a more specific recursive failure. Of the 21 complete trajectories obtained from seven throat velocities and three seeds, one prediction left the timelike region. The exit occurred for $u_{\text{th}} = 0.05$, and the other predictions for this family reached the throat with appreciably different velocities.

The failure is consistent with the geometry already visible in Figure 2. In the incoming region, different conserved-energy branches may have almost the same (x, ξ) coordinates and very similar changes over one short interval, although they later separate strongly near the throat. A model supplied only with the local state must recover the trajectory family from these small differences at every recursive step.

The conserved energy provides an exact label for that family. Although $\mathcal{E}_0$ is determined by an exact state and therefore adds no new physical information, it need not be represented with sufficient accuracy by a finite network trained only on local increments. Supplying it explicitly asks a simpler question of the model: predict the local motion at (x, ξ) for the specified conserved-energy branch. During a recursive trajectory, $\mathcal{E}_0$ was calculated once from the exact initial state and held fixed. The energy calculated from the predicted state was used only as a diagnostic and was not fed back into the model.

The controlled comparison used

$$(x, \xi) \mapsto (\Delta x, \Delta \xi), \quad 2 \rightarrow 32 \rightarrow 32 \rightarrow 2,$$

for the model without explicit energy, and

$$(x, \xi, \mathcal{E}_0) \mapsto (\Delta x, \Delta \xi), \quad 3 \rightarrow 32 \rightarrow 32 \rightarrow 2,$$

when the conserved energy was included. This is a later matched comparison and should not be treated as a fourth entry in Table 2. Apart from the additional input, the two models used the same 40,000 training states, the same 8,000 validation states, the same target definition and training-only standardization, and the same three seeds. Both used two tanh hidden layers, the same standardized mean-squared error for the two outputs, and the same optimization and stopping protocol.

The one-step results changed only modestly and were not uniformly better. On the common validation set, the mean $\Delta \xi$ RMSE decreased from 3.3723×10^{-5} to 3.1951×10^{-5} , a reduction of 5.3%, while the mean Δx RMSE increased from 1.1732×10^{-4} to 1.3319×10^{-4} . The recursive trajectories changed much more strongly, particularly for the three lowest throat velocities. Table 3 gives the aggregate quantities that are most useful for separating these two effects.

Table 3. Local and recursive results without and with the fixed conserved-energy input.

Quantity	Without explicit energy	With fixed $\mathcal{E}_0$
Validation Δx RMSE	1.1732×10^{-4}	1.3319×10^{-4}
Validation $\Delta \xi$ RMSE	3.3723×10^{-5}	3.1951×10^{-5}
Mean absolute recursive ξ error, nine difficult incoming rollouts	2.6952×10^{-2}	1.4835×10^{-3}
Full trajectories reaching $x = 17$	20 / 21	21 / 21
Physical exits	1 / 21	0 / 21
Mean maximum pre-throat relative energy drift	0.10791	0.01089

The nine difficult incoming cases consist of the three seeds for $u_{\text{th}} = 0.05, 0.15$, and 0.30 . Across them, supplying the fixed energy reduced the mean absolute recursive ξ error by 94.5%. The mean absolute error in the recovered throat velocity decreased from 0.02814 to 0.01493 for $u_{\text{th}} = 0.05$, from 0.01116 to 0.00335 for $u_{\text{th}} = 0.15$, and from 0.00531 to 0.00127 for $u_{\text{th}} = 0.30$. All three seeds then completed every one of these difficult traversals. Across all seven trajectory families, the single physical exit of the model without explicit energy was removed, and the maximum energy drift accumulated before the throat was also substantially reduced.

Figure 4 shows the fixed-energy predictions for the hardest trajectory in both physical and transformed coordinates. All three seeds remain inside the timelike region and complete the traversal. The transformed view also shows that the agreement is not exact: the predicted curves remain close over most of the incoming branch and separate visibly from one another near the throat.

The improvement was not uniform over the full family. For $u_{\text{th}} = 0.80$ and 0.90 , the throat-velocity errors increased slightly, although they remained of order 10^{-3} , and the hardest $u_{\text{th}} = 0.05$ trajectory was still not recovered exactly. Throat-started controls reached $x = 17$ for every seed with and without the energy input. This indicates that the main gain came from preserving the trajectory family through the compressed incoming branch rather than from changing the already well-represented outgoing motion.

The conserved energy was therefore retained because it improved the recursive preservation of physically distinct trajectory families, not because it uniformly reduced every local error. Its effect also makes the distinction between local and recursive evaluation more concrete: a small change in the one-step averages accompanied a much larger change in the trajectories obtained after repeated application. The remaining throat error and seed dependence show that explicit energy does not by itself solve the limitations of recursive local prediction, but it removes one important ambiguity in the compressed incoming state.

The remaining error suggested a second question: whether a loss that is more closely aligned with energy preservation or with the sensitivity of the future throat crossing could improve the complete trajectory. These tests kept the fixed-energy inputs, architecture, data split, optimizer, stopping rule, and three seeds unchanged. Let e be the two-component standardized increment error and let n_E be the unit normal obtained from the local energy gradient in the same standardized coordinates. The energy-normal loss was

$$\mathcal{L}_{\text{EN}} = \mathcal{L}_{\text{MSE}} + \frac{1}{2N} \sum_{k=1}^N (n_{E,k} \cdot e_k)^2.$$

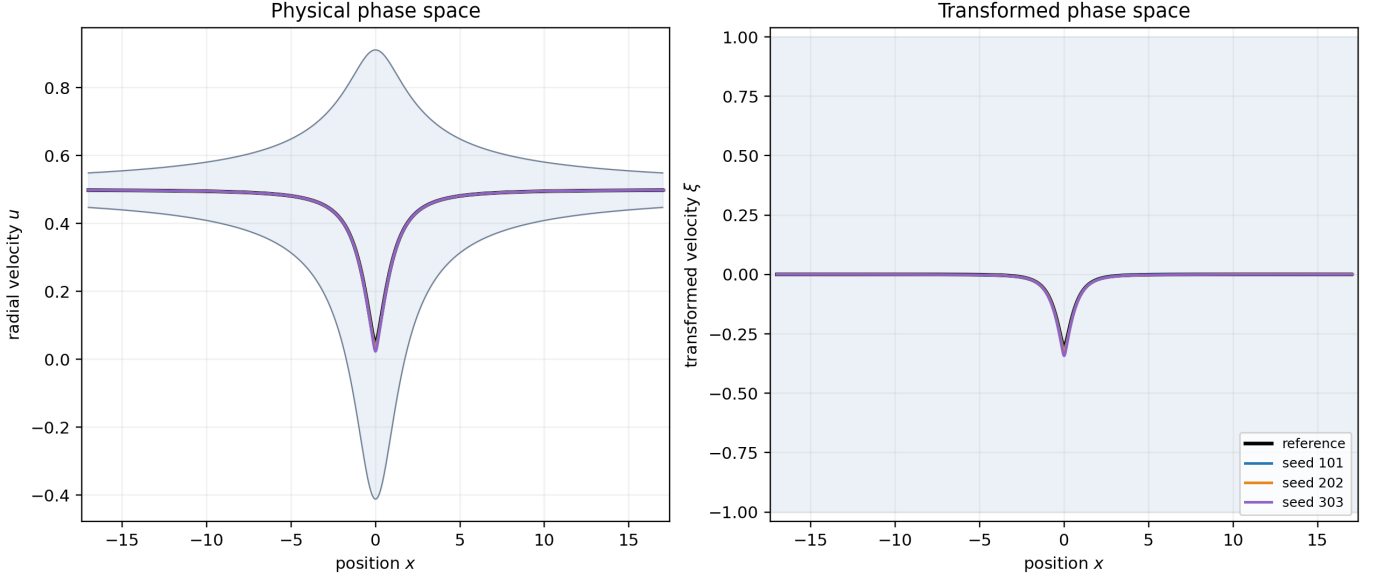


Figure 4: Exact and recursively predicted trajectories for $u_{\text{th}} = 0.05$ with $\mathcal{E}_0$ supplied as a fixed input. The left panel shows the physical (x, u) phase space and its timelike region; the right panel shows the same trajectories in (x, ξ) , the representation used by the model. All three seeds complete the traversal, although the remaining spread near the throat is visible in both coordinates.

It reduced the validation root-mean-square energy-normal error by 14.2% and the independently evaluated nonlinear one-step energy-displacement MAE by 22.7%. This local improvement did not persist through the most difficult recursive trajectory: the mean throat-velocity error for $u_{\text{th}} = 0.05$ increased from 0.01493 to 0.03919, and one of the 21 full rollouts again crossed the timelike boundary.

A further loss concentrated the same energy-normal error on states whose perturbations have a large effect on the future throat velocity. With I_k selecting eligible incoming states and w_k denoting the normalized absolute sensitivity, the added term was

$$\mathcal{L}_S = \mathcal{L}_{\text{MSE}} + \frac{1}{2N} \sum_{k=1}^N I_k w_k (n_{E,k} \cdot e_k)^2.$$

The standardized MSE in the targeted micro-core decreased by 49.5%, and the energy-normal RMS error far upstream on the $u_{\text{th}} = 0.05$ trajectory decreased by 43.2%. Even so, the corresponding throat-velocity error increased to 0.01607 and the same aggregate completion count, 20 successful traversals and one physical exit, was obtained. Table 4 places the local gains beside these recursive outcomes.

Table 4. Controlled local-loss experiments. Each treatment is compared with the same fixed-energy baseline.

Treatment	Directly targeted local change	Additional local diagnostic	$u_{\text{th}} = 0.05$ throat MAE	Full reach / exit
Fixed-energy baseline	—	—	0.01493	21 / 0
Energy-normal loss	energy-normal RMS: −14.2%	nonlinear one-step energy MAE: −22.7%	0.03919	20 / 1
Sensitivity-aware loss	targeted micro-core MSE: −49.5%	far-upstream energy-normal RMS: −43.2%	0.01607	20 / 1

These failures are useful because they rule out a simple explanation in terms of an incorrectly chosen one-step norm. Both losses moved their intended local quantities in the desired direction, in some cases quite strongly, but the signed errors accumulated along the incoming branch were not controlled well enough to improve the eventual crossing. The next part of the study therefore changes the prediction problem itself. Instead of learning a small step and composing it many times, we ask for the state reached after the complete requested time interval in one evaluation.

4. Direct finite-time prediction

The direct problem begins with complete exact orbits. A throat velocity u_{th} fixes the conserved energy, from which the state on the incoming boundary $x = -17$ is recovered. The exact equations are then integrated until the particle reaches $x = 17$. From any state (x_0, ξ_0) on such an orbit and any non-negative elapsed time s that remains within the stored traversal, the desired map is

$$\Psi_s : (x_0, \xi_0, \mathcal{E}_0, s) \mapsto (x_s, \xi_s).$$

The training, validation, and held-out banks contain 4,096, 1,024, and 1,024 complete orbits, respectively, with the split made at the level of the parent orbit. The seven named trajectories shown in Figure 2 were stored separately and used only as reference cases. Each parent orbit supplied 96 transitions: 64 rows distributed over the full orbit and 32 additional rows used either to resolve the difficult low- u_{th} families or to provide a second independent global sample. This gave 393,216 training rows and 98,304 rows in each of the validation and held-out sets. The difficult class was defined in advance by $u_{\text{th}} \leq 0.30$, giving 2,231 training, 558 validation, and 557 held-out difficult orbits. Because every transition from a given orbit belongs to only one split, two rows cannot share a trajectory segment across training and evaluation.

Identity examples were included deliberately rather than being left for the network to infer from nearby short-time samples. The training and validation sets together contain 20,480 rows with $s = 0$ and exactly zero state increment. The remaining times range from very short intervals to the endpoint of the complete traversal. The 32 additional rows on each difficult orbit emphasize the compressed incoming region, its exit, the central passage, and long predictions to the outgoing branch. The sampled data therefore represent both the ordinary part of the orbit family and the transitions that were least reliable under recursive local prediction.

The input is

$$q = (x_0, \xi_0, \mathcal{E}_0, s),$$

and the common network has two tanh hidden layers,

$$4 \longrightarrow 64 \longrightarrow 64 \longrightarrow 2.$$

In the notation introduced in Section 3, its parameter shapes are

$$W^{(1)} \in \mathbb{R}^{64 \times 4}, \quad W^{(2)} \in \mathbb{R}^{64 \times 64}, \quad W^{(3)} \in \mathbb{R}^{2 \times 64},$$

with biases in $\mathbb{R}^{64}$, $\mathbb{R}^{64}$, and $\mathbb{R}^2$. The model consequently has 4,610 trainable parameters. Inputs and targets are again standardized using training data only, and the final layer remains linear. Training used Adam with learning rate 10^{-3} , batches of 512 rows, float32 arithmetic, zero weight decay, no learning-rate scheduler, a maximum of 1,500 epochs, and early-stopping patience 40.

The representation of the two output targets required a separate comparison. Directly learning the accumulated increments,

$$\Delta x = x_s - x_0, \quad \Delta \xi = \xi_s - \xi_0,$$

asks one network output to approach zero as s approaches zero while also covering the displacement of a complete traversal. It does not make Ψ_0 exact. Dividing by elapsed time gives average-rate targets,

$$V_x = \frac{\Delta x}{s}, \quad V_\xi = \frac{\Delta \xi}{s},$$

with the exact analytic limits $V_x(0) = u_0$ and $V_\xi(0) = \dot{\xi}_0$. At inference, multiplication by the physical elapsed time makes both reconstructed increments exactly zero when $s = 0$. This formulation greatly improves the short-time limit, but it also asks the ξ output to describe one average rate over intervals whose lengths vary from almost zero to the complete orbit.

The selected representation keeps the ordinary rate for x but uses a nonlinear time factor for ξ ,

$$V_x = \frac{\Delta x}{s}, \quad F_\xi = \frac{\Delta \xi}{g(s)}, \quad g(s) = -5 \exp m1 \left(-\frac{s}{5} \right).$$

The predicted increments are reconstructed as

$$\widehat{\Delta x} = s\widehat{V}_x, \quad \widehat{\Delta \xi} = g(s)\widehat{F}_\xi.$$

Here the factors s and $g(s)$ are applied in physical dimensionless time, after the network calculation; they are not standardized inputs. The nonlinear factor satisfies

$$g(0) = 0, \quad g'(0) = 1, \quad g(s) = s - \frac{s^2}{10} + O(s^3), \quad \lim_{s \rightarrow \infty} g(s) = 5.$$

It therefore has the same first-order behaviour as elapsed time near the identity, while avoiding indefinite growth of the scale used for the bounded ξ displacement. We refer to this as the time-rescaled finite-time model.

Table 5 compares the three representations on the frozen validation transitions. A separate model and checkpoint were selected for each formulation. The comparison shows why neither identity alone nor aggregate error alone was sufficient. The pure rate produces exact identity and the smallest short-time errors, but its long-time ξ error is larger than that of the accumulated-increment model. The time-rescaled target retains exact identity and lowers the pure-rate ξ RMSE by 22.4% at intermediate times and 36.0% at long times, while also giving the smallest aggregate x and ξ errors of the three.

Table 5. Validation RMSE for the three finite-time target representations.

Representation	Exact identity	Regime	x RMSE	ξ RMSE
Accumulated increment	no	all rows	0.126167	0.00344346
Average rate	yes	all rows	0.0699973	0.00471531
Time-rescaled	yes	all rows	0.0667078	0.00308242
Accumulated increment	no	$s \leq 5$	0.0542956	0.00128671
Average rate	yes	$s \leq 5$	0.000969386	0.000261692
Time-rescaled	yes	$s \leq 5$	0.00141286	0.000427107
Accumulated increment	no	$5 < s \leq 20$	0.0635822	0.00224363
Average rate	yes	$5 < s \leq 20$	0.0126727	0.00240533
Time-rescaled	yes	$5 < s \leq 20$	0.0121616	0.00186701
Accumulated increment	no	$s > 20$	0.202146	0.00539541
Average rate	yes	$s > 20$	0.120664	0.00782605
Time-rescaled	yes	$s > 20$	0.114980	0.00500568

The retained time-rescaled model was seed 202, chosen only by the orbit-averaged standardized target MSE on the validation orbits. Its best epoch was 569 and training stopped at epoch 609. No physical trajectory metric was used to select this seed or checkpoint. On a denser grid of 524,288 validation queries, it gave RMSE values 0.063681 in x and 0.00295434 in ξ , with no prediction outside $-1 < \xi < 1$. Figure 5 shows direct predictions from three starting positions on the most difficult named reference trajectory. The agreement is close on the scale of the complete orbit, although the residual displacement becomes more visible for long predictions from the incoming branch.

The direct model and the local model answer different questions. At $s = 0.2$, the direct model evaluates the requested finite-time map once, whereas the local model performs its native one-step prediction. At longer times, the direct model still makes one prediction from the exact anchor, while the local model is applied recursively in increments of 0.2. We compared them on eighteen fixed anchors from each of the 1,024 validation orbits and each of the seven named reference trajectories, giving 18,558 matched starting states. The local comparator was the fixed-energy seed-101 model from Section 3.

At the native interval, the local model remains clearly more accurate. Its dense matched RMSE is 1.13105×10^{-4} in x and 1.74116×10^{-5} in ξ , compared with 1.53120×10^{-4} and 6.17314×10^{-5} for direct prediction. Thus the direct-to-local error ratios are 1.35 and 3.55. This is the expected advantage of fitting one fixed, short map. The distinction changes as the local errors are repeatedly returned to the model as new inputs. Figure 6 illustrates the comparison from two incoming anchors on the difficult reference trajectory.

Across the validation comparison, the direct model first has a smaller x RMSE at $s = 0.8$, although isolated early crossings of the two error curves are not stable. Its x RMSE remains lower from $s = 22$ onward, and its x MAE remains lower from $s = 24$ onward. No corresponding sustained global crossover occurs for ξ through $s = 60$. The population being compared becomes smaller at late times because a recursive local trajectory is included only while both predictions remain available and physical: the common count decreases from 18,558 at $s = 0.2$ to 3,198 at $s = 60$. Figure 7 therefore reports the common-survivor error together with this evaluation convention; it should not be interpreted as an unconditional error over all initial anchors.

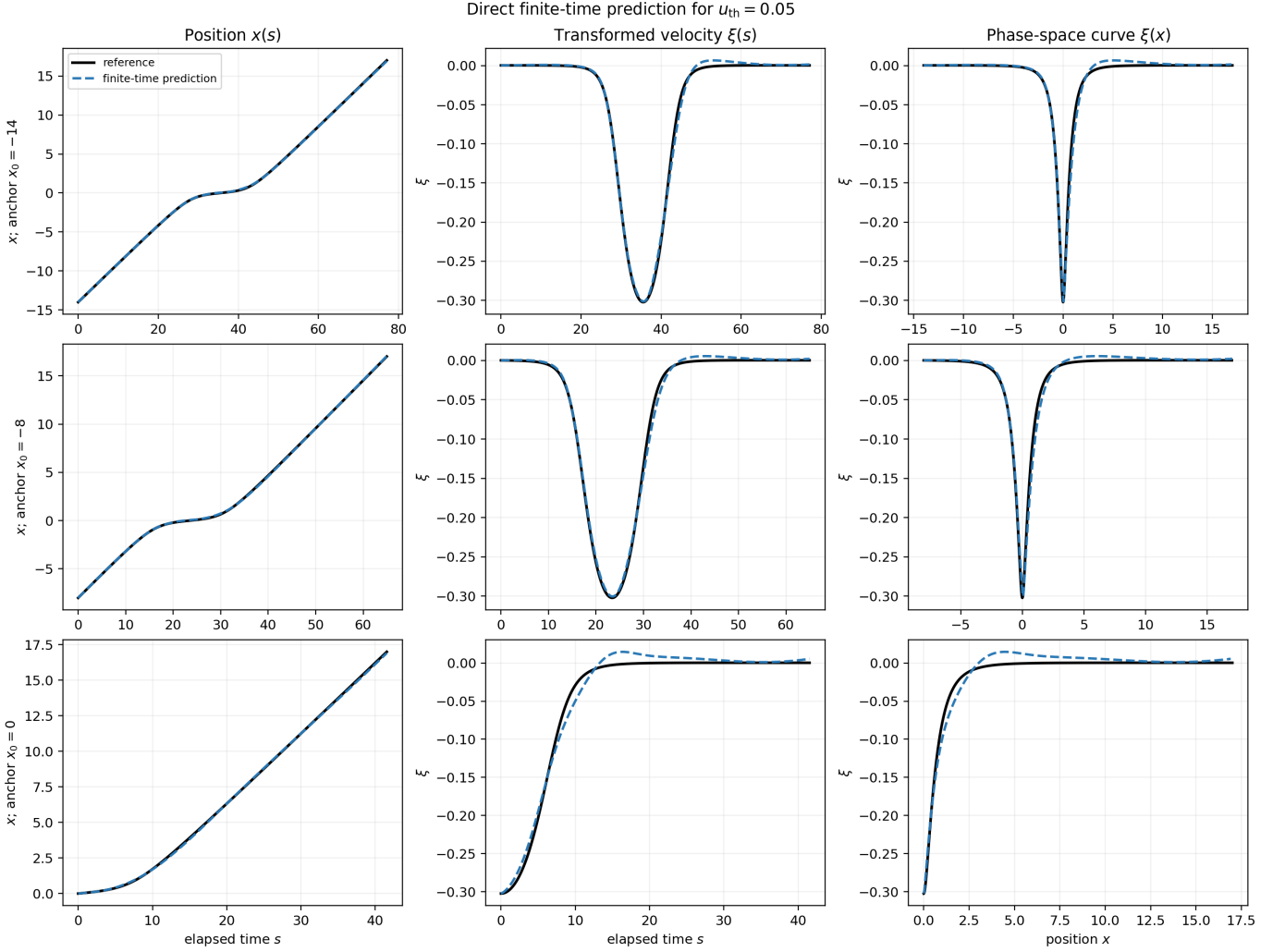


Figure 5: Exact and direct finite-time predictions for the named $u_{th} = 0.05$ trajectory, starting from $x_0 = -14$, -8 , and 0 . The upper and middle rows show $x(s)$ and $\xi(s)$; the lower row shows the corresponding paths in transformed phase space. The reference trajectory is separate from the validation aggregates.

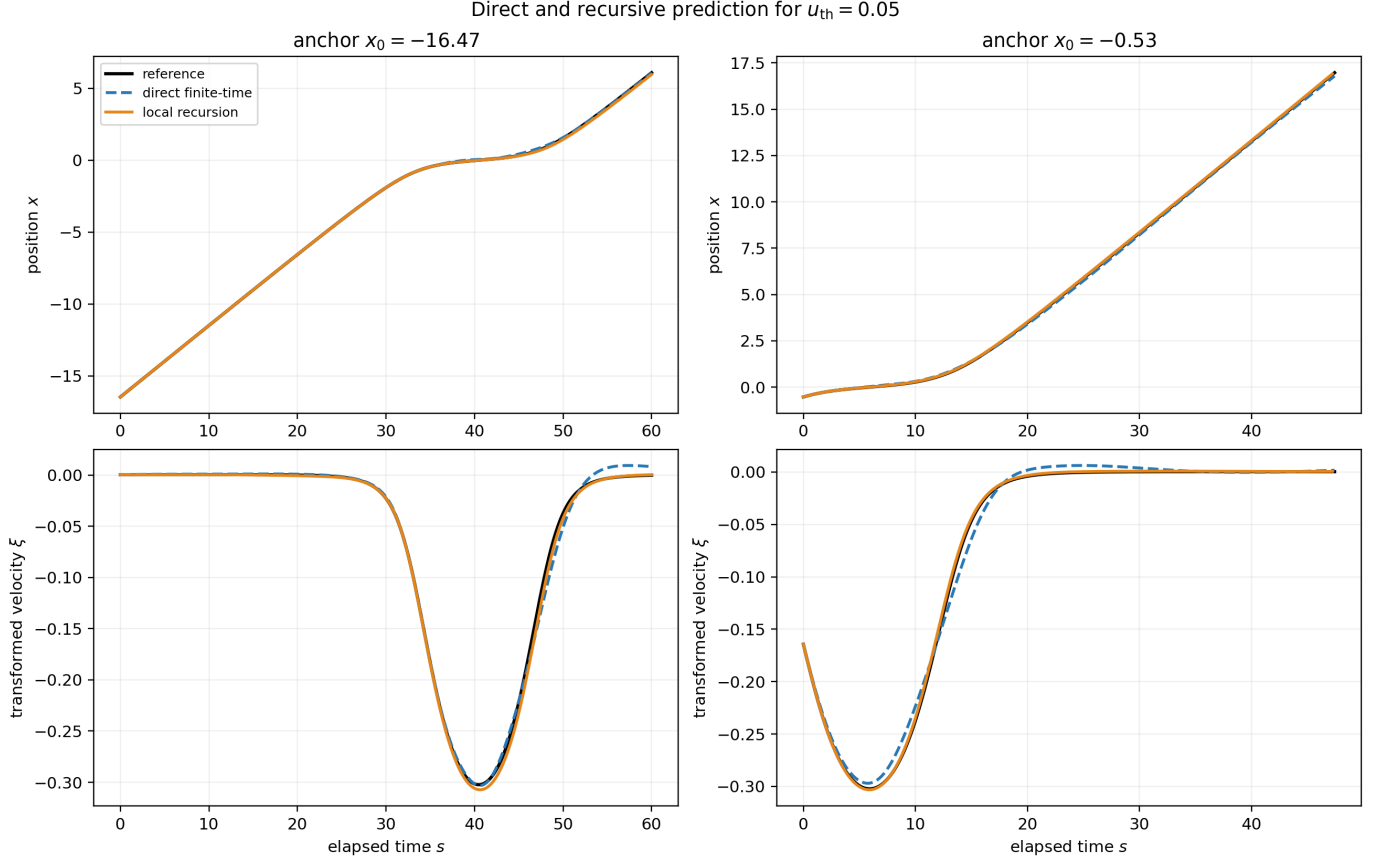


Figure 6: Exact motion, one-shot finite-time prediction, and recursive local prediction for $u_{th} = 0.05$, shown from two incoming starting positions. The direct curve is evaluated independently at each displayed time from the exact anchor. The local curve is obtained by repeated $h = 0.2$ residual updates.

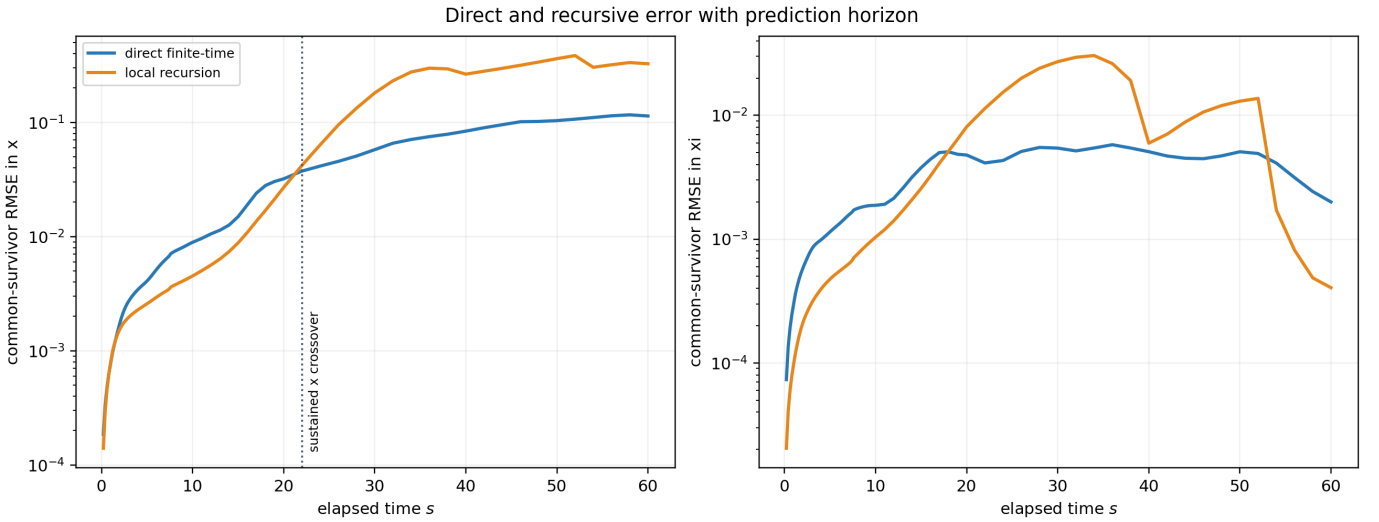


Figure 7: Validation RMSE in x and ξ for direct finite-time prediction and the recursively applied local model, evaluated on states for which both predictions survive. The vertical line marks the sustained x -RMSE crossover at $s = 22$. The changing common-survivor population is accounted for in the reported comparison.

The direct model therefore does not replace the local map at its native scale. It trades some short-time accuracy for the ability to reach a distant time without repeatedly feeding approximation error back into the state. For x , that trade becomes favourable over sufficiently long traversals. For ξ , the local model retains the lower validation RMSE among the surviving trajectories, even though its recursive state can develop the phase and family errors discussed earlier. These are complementary statements, and both are needed to describe the two prediction strategies fairly.

5. Structure of the remaining finite-time error

The aggregate finite-time errors conceal a pronounced dependence on the trajectory family. On the dense validation grid, the time-rescaled seed-202 model has x and ξ RMSE values 0.0846704 and 0.00386752 for $u_{\text{th}} \leq 0.30$, compared with 0.0180755 and 0.00112636 for the remaining orbits. The difficult-to-ordinary ratios are therefore about 4.68 in x and 3.43 in ξ . The largest errors also occur more often for long elapsed times and for anchors on the compressed incoming branch. This is consistent with the geometry of the exact orbit family, but it does not yet specify which part of the motion the network fails to reproduce.

To resolve the rapid central change without selecting regions from model errors, we used the exact transformed velocity. A rapid-change feature was defined by

$$|\dot{\xi}| > 0.036514132,$$

where the threshold is the 95th percentile of $|\dot{\xi}|$ in the training dynamics. Threshold crossings were interpolated and nearby intervals were merged before any derivative-loss model was trained. The resulting validation set contains 857 rapid-change features and 3,716, 5,455, and 7,387 rows in matched pre-change, during-change, and post-change regions. This construction is based only on the exact dynamics and gives a fixed way to ask whether an error in the rapid part of the orbit is followed by a larger state error.

The time-rescaled representation permits the physical-time derivative of the predicted ξ to be calculated directly. Since

$$\hat{\xi}(s) = \xi_0 + g(s)\widehat{F}_\xi(q),$$

its derivative includes both the explicit gate and the dependence of the network output on its time input,

$$\frac{\partial \hat{\xi}}{\partial s} = g'(s)\widehat{F}_\xi + g(s)\frac{\partial \widehat{F}_\xi}{\partial s}.$$

The controlled derivative loss was

$$\mathcal{L} = \mathcal{L}_{\text{state}} + \frac{\lambda_{\dot{\xi}}}{N} \sum_{k=1}^N \left(\frac{\partial_s \hat{\xi}_k - \dot{\xi}_k}{\sigma_{\dot{\xi}}} \right)^2,$$

where $\sigma_{\dot{\xi}}$ is calculated from the training values. The architecture, input and target preprocessing, data rows, optimizer, stopping rule, and seeds were otherwise unchanged. The zero-weight runs reproduced the retained baseline checkpoints, so the comparison isolates the added derivative term.

The principal results are given in Table 6. The intermediate weight $\lambda_{\dot{\xi}} = 0.034$ reduces the error in $\dot{\xi}$ during the rapid features by 20.2%. Its effect on the state after the feature is smaller, but the global ξ error and the long-time difficult-family ξ error also decrease. The larger weight 0.068 gives almost the same rapid-feature improvement while increasing the global x RMSE by 7.3%. The value 0.034 was therefore retained as the better validation tradeoff, rather than as a universally optimal physical coefficient.

Table 6. Across-seed validation means for the physical-time derivative-loss experiment. Parentheses give the change from $\lambda_{\dot{\xi}} = 0$.

$\lambda_{\dot{\xi}}$	During-feature $\dot{\xi}$ RMSE	Post-feature ξ RMSE	Global ξ RMSE	Global x RMSE	Difficult, $s > 20$, ξ RMSE
0	0.00326700	0.00627713	0.00349921	0.0757273	0.00717919
0.011	0.00288954 (−11.6%)	0.00595006 (−5.2%)	0.00332575 (−5.0%)	0.0721751 (−4.7%)	0.00684203 (−4.7%)
0.034	0.00260759 (−20.2%)	0.00613961 (−2.2%)	0.00317748 (−9.2%)	0.0749630 (−1.0%)	0.00658232 (−8.3%)

$\lambda_{\dot{\xi}}$	During-feature $\dot{\xi}$ RMSE	Post-feature ξ RMSE	Global ξ RMSE	Global x RMSE	Difficult, $s > 20$, ξ RMSE
0.068	0.00261739 (−19.9%)	0.00591980 (−5.7%)	0.00332392 (−5.0%)	0.0812553 (+7.3%)	0.00677369 (−5.7%)

The reduction in the post-feature coordinate error is modest compared with the reduction in the derivative error itself. We therefore decomposed the remaining phase-space displacement into components tangent and normal to the exact orbit. Let

$$\tilde{e} = \left(\frac{\hat{x} - x}{\sigma_x}, \frac{\hat{\xi} - \xi}{\sigma_{\xi}} \right), \quad \sigma_x = 10.2010437605, \quad \sigma_{\xi} = 0.206945372604,$$

and let τ be the unit tangent of the exact trajectory in these scaled coordinates. We define

$$e_{\parallel} = \tilde{e} \cdot \tau, \quad e_{\perp} = \|\tilde{e} - e_{\parallel} \tau\|_2.$$

In the lowest throat-velocity bin, $0.01 \leq u_{\text{th}} \leq 0.05$, the control model has post-change RMSE 0.102904 in the tangent direction and 0.040551 in the normal direction. The tangent error is therefore 2.54 times larger. Adding the selected derivative loss reduces the normal component to 0.037850, a 6.65% improvement reproduced by all three seeds, but leaves the tangent component essentially unchanged at 0.103004. The network is thus often close to the correct orbit while being displaced along it, which is a phase or timing error rather than primarily a departure from the physical trajectory family.

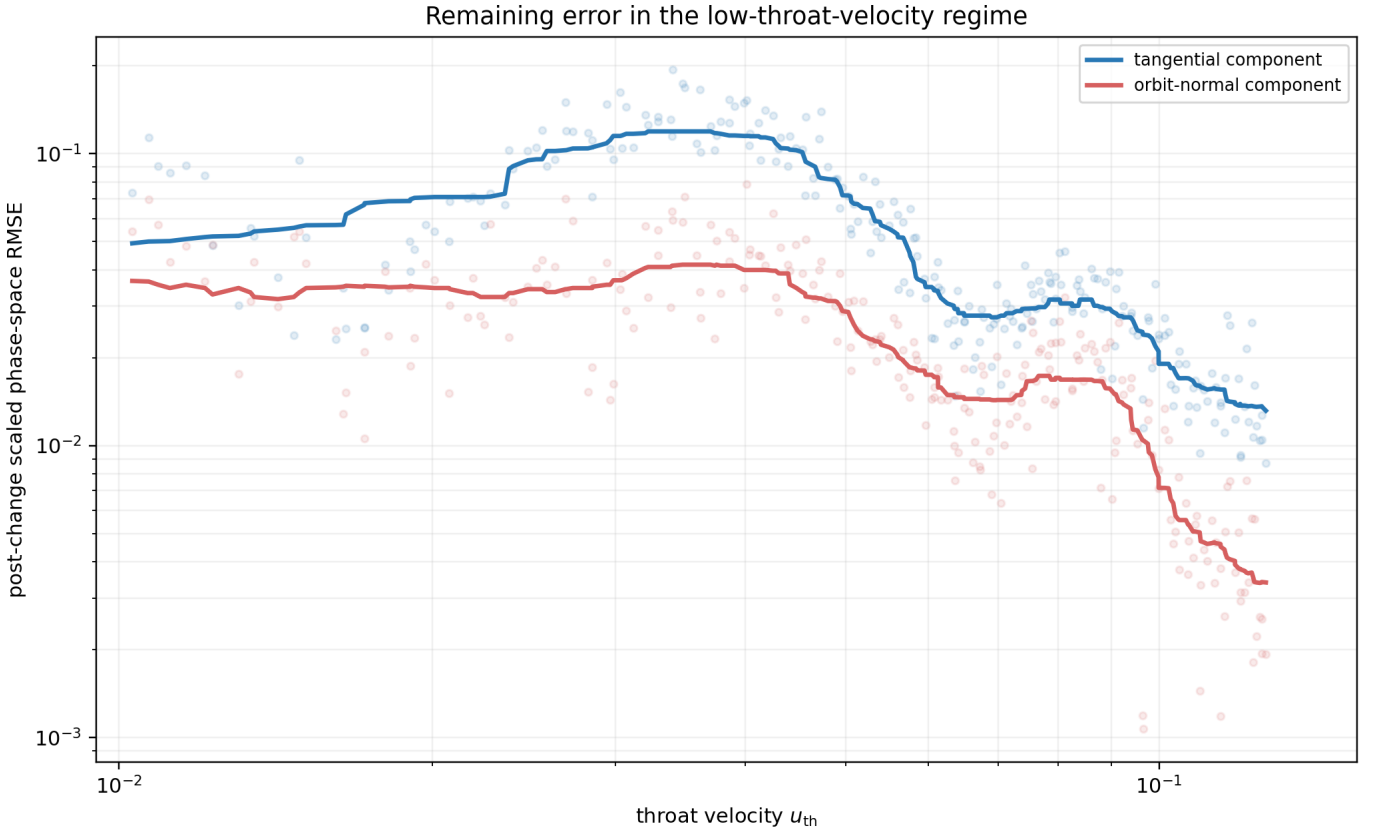


Figure 8: Post-change phase-space error for the control finite-time models, decomposed into scaled tangent and orbit-normal components and plotted against u_{th} . Points show orbit-level values from the three seeds and curves show rolling medians. Continuous post-change coverage exists only through $u_{\text{th}} = 0.15$, so the figure stops there rather than extrapolating across an unsampled interval.

This decomposition uses 252 distinct validation orbits over the continuously covered interval $u_{\text{th}} \leq 0.15$. There are no post-change samples in the next interval $0.15 < u_{\text{th}} \leq 0.30$, and no claim is made about a smooth continuation across that gap. The derivative and phase-space analyses are also validation studies. They explain part of the remaining error and support a possible training modification, but they are not part of the sealed evaluation of the originally selected seed-202 model.

6. Frozen held-out evaluation

The final held-out evaluation was performed only after the finite-time target representation, preprocessing, selected shared architecture, seed-202 checkpoint, query grids, local comparator, metrics, and crossover definitions had been frozen. The selected checkpoint was not retrained or adjusted. The held-out set contains 1,024 complete orbits and 98,304 stored transitions; the dense evaluation contains 524,288 direct queries, the native-step comparison contains 32,768 matched queries, and the direct-versus-recursive comparison contains 18,432 exact anchors. The seven named reference trajectories are not included in any held-out aggregate.

Table 7 compares the validation and held-out results. The stored-row and dense errors reproduce closely, including within the difficult family. Exact identity is retained, and none of the dense direct predictions violates $-1 < \xi < 1$ or the equivalent timelike condition. The dense energy MAE, which is a diagnostic rather than a trained loss, also changes very little.

Table 7. Frozen validation and held-out evaluation of the selected time-rescaled seed-202 model.

Quantity	Validation	Held-out test
Stored-row x RMSE	0.0667078	0.0661458
Stored-row ξ RMSE	0.00308242	0.00309810
Dense x RMSE	0.0636810	0.0629547
Dense ξ RMSE	0.00295434	0.00293201
Difficult-family stored-row x RMSE	0.0890080	0.0883275
Difficult-family stored-row ξ RMSE	0.00403327	0.00406086
Dense energy MAE	0.00477153	0.00473772
Exact identity residual in (x, ξ)	(0, 0)	(0, 0)
Dense admissibility violations	0	0
Direct/local RMSE ratio at $s = 0.2$, x	1.35	1.35
Direct/local RMSE ratio at $s = 0.2$, ξ	3.55	3.53
Sustained global x -RMSE crossover	$s = 22$	$s = 24$

The agreement is not limited to one aggregate number. The held-out hard-to-ordinary RMSE ratios are 4.64 in x and 3.41 in ξ , close to the corresponding validation structure. The local model remains better at $s = 0.2$, with held-out local RMSE values 1.13042×10^{-4} and 1.74344×10^{-5} , compared with 1.52817×10^{-4} and 6.15776×10^{-5} for the direct model. Conversely, the sustained global x -RMSE crossover moves only from $s = 22$ in validation to $s = 24$ in the held-out cohort; the sustained x -MAE crossover is $s = 24$ in both.

One result does not reproduce in the same simple way. The held-out global common-survivor comparison shows a sustained ξ -RMSE crossover at $s = 50$, whereas validation showed no sustained crossover through $s = 60$. This late result is absent in the ordinary-family comparison and is not reproduced by the separately examined near- u_{th} families. It is therefore best read as a property of the surviving held-out cohort at late time, not as robust evidence that the direct model has generally become more accurate in ξ .

The held-out calculation supports the main finite-time results without removing their qualifications. The direct model reproduces its validation accuracy and physical admissibility on unseen complete orbits. It is still less accurate than the local model over one native step, and its advantage over recursive prediction is clearest for the accumulated x error at long times. The derivative-loss and output-splitting studies were not tested on this sealed cohort, because doing so after inspecting their validation results would have changed the role of the test set.

7. Output specialization and the split-head test

The two finite-time outputs do not behave identically. The x target benefits strongly from rate scaling and shows a clear long-time advantage over recursive prediction, while the bounded ξ target requires a saturating time factor and retains a concentrated low-energy error. This raised the possibility that a completely shared hidden representation was forcing an unhelpful compromise between them.

The retained shared network may be written as

$$h^{(1)} = \tanh(W^{(1)}\tilde{q} + b^{(1)}), \quad h^{(2)} = \tanh(W^{(2)}h^{(1)} + b^{(2)}),$$

$$(\widehat{V}_x, \widehat{F}_\xi) = W^{(3)}h^{(2)} + b^{(3)},$$

with widths $4 \rightarrow 64 \rightarrow 64 \rightarrow 2$ and 4,610 parameters. The tested alternative kept one shared 64-unit layer and then introduced two separate 32-unit branches,

$$h_x = \tanh(W_x h^{(1)} + b_x), \quad h_\xi = \tanh(W_\xi h^{(1)} + b_\xi),$$

$$\widehat{V}_x = w_x^\top h_x + c_x, \quad \widehat{F}_\xi = w_\xi^\top h_\xi + c_\xi.$$

This $4 \rightarrow 64 \rightarrow (32, 32) \rightarrow (1, 1)$ late-split network has 4,546 parameters. It is slightly smaller than the shared model and should be regarded as parameter-count-controlled rather than exactly parameter-matched. The dataset, preprocessing, targets, identity gates, derivative normalization, optimizer, epoch limit, early stopping, seed-specific shuffling, and checkpoint criterion were unchanged.

Table 8 gives the primary comparison at the selected derivative weight $\lambda_\xi = 0.034$. The split network lowers the global x RMSE by 2.14%, with the same direction of change for all three seeds. It does not improve the quantities that motivated the split: global ξ , difficult long-time ξ , post-change ξ , and the low-energy orbit-normal error all increase. The last two increases have the same sign for every seed. Energy consistency also becomes slightly worse on average.

Table 8. Shared and late-split validation results at $\lambda_\xi = 0.034$. Relative changes are matched-seed changes from shared to split.

Metric	Shared	Late split	Relative change	Seed consistency
Global x RMSE	0.0749630	0.0733971	−2.14%	improvement in all seeds
Global ξ RMSE	0.00317748	0.00335826	+5.56%	mixed
Difficult, $s > 20$, ξ RMSE	0.00658232	0.00687327	+4.27%	mixed
Post-change ξ RMSE	0.00613961	0.00647643	+5.53%	degradation in all seeds
Lowest-bin post-change $e_\perp$ RMSE	0.0378500	0.0414972	+9.85%	degradation in all seeds
Lowest-bin post-change $e_\parallel$ RMSE	0.103004	0.105074	+2.09%	degradation in all seeds
Energy RMSE	0.0208679	0.0217206	+10.79%	mixed

The single dense $u_{\text{th}} = 0.05$ reference is more favourable to the split network, including a smaller orbit-normal error. That isolated curve does not represent the frozen low-energy population: over the complete lowest bin, the defining normal residual becomes larger rather than smaller. This difference is a useful warning against selecting an architecture from one visually convincing trajectory.

The result does not show that output specialization is generally unsuitable for this problem. It shows only that separating the final hidden layer into two equal, relatively small branches does not improve the residual for which it was introduced. Earlier separation, unequal branch capacities, a smaller shared trunk with larger branches, or completely separate networks remain different hypotheses. For example, a $4 \rightarrow 32 \rightarrow (48, 48) \rightarrow (1, 1)$ model would move more capacity into the specialized branches, but this architecture has not been trained and is not part of the evidence reported here.

8. Discussion

The progression of the experiments is easier to understand if the learned maps are separated by what they are required to preserve. A local model approximates a short flow map accurately on exact states. A recursively used local model must also remain accurate on its own slightly displaced states and must preserve small differences between neighbouring trajectory families over many steps. A direct finite-time model has a third task: it must represent a continuum of maps from the identity to almost a complete traversal. These requirements overlap, but the results show that none of them can be inferred reliably from another.

The first useful change was physical rather than architectural. Mapping the position-dependent timelike interval to $-1 < \xi < 1$ removes a known geometrical variation from the input domain. Supplying $\mathcal{E}_0$ then identifies the exact orbit family through the part of phase space where different incoming trajectories are strongly compressed. The energy is formally determined by an exact state, so it does not add a new degree of freedom to the dynamics. It nevertheless changes the numerical learning problem, because a finite network no longer has to recover the conserved branch from small local state differences at every recursive step. The much larger improvement in complete rollouts than in one-step RMSE is consistent with this interpretation.

The unsuccessful local-loss experiments refine that conclusion. They show that reducing an energy-normal norm, even when it is concentrated on the most sensitive incoming states, is not enough to control the signed sequence of errors that determines the later throat crossing. The difference between absolute local improvement and coherent cumulative bias is particularly relevant in low-dimensional dynamics, where a small error may persist in nearly the same direction over a long part of an orbit. A loss can therefore improve the distribution it was designed to measure and still produce a worse recursive trajectory.

Direct finite-time prediction avoids this particular accumulation by returning every requested state to the same exact anchor. The choice of target representation is then part of the mathematical model, not merely a preprocessing detail. Multiplying learned rate-like outputs by functions that vanish at $s = 0$ builds the identity condition into the prediction. The linear factor s is natural for x , which accumulates throughout the motion, while the saturating factor $g(s)$ better matches the bounded displacement of ξ . This construction gives exact identity without constraining the network weights, and it explains why the time-rescaled model can behave differently from a network trained on the raw increments even when the architecture and data are unchanged.

The direct model’s long-time advantage is confined mainly to x . The orbit-level decomposition helps explain why. In the lowest-energy population, most of the remaining scaled phase-space error lies along the exact trajectory rather than normal to it. A prediction can therefore remain near the correct orbit and satisfy the timelike constraint while arriving too far ahead or behind along that orbit. Derivative supervision reduces part of the smaller normal component, but it does not substantially reduce the dominant tangential component. This also explains why zero admissibility violations and small energy error, although necessary physical checks, do not by themselves imply an accurate finite-time phase.

The late split-head result gives a similar warning about architectural interpretation. The two outputs have different statistical behaviour, but separating only the last hidden representation does not reduce the low-energy residual over the validation population. A single reference trajectory suggests the opposite conclusion, which is why population-level and seed-level comparisons were retained. The experiment rules out one specific late split; it does not show that the two outputs should always share all of their representation.

Taken together, the results favour a layered use of physical information. Known domain geometry is best handled by an exact coordinate transformation, the conserved branch can be supplied as a fixed input when local states are compressed, and exact identity can be imposed through the finite-time output form. Loss terms and architectural changes are then judged by whether they improve the trajectory-level quantity that motivated them, not only the local term they directly optimize.

9. Limitations and future directions

All calculations in this report use one wormhole and spiral configuration, with $m = 2$, $\omega = 1$, $\alpha = -2$, and $\theta = \pi/6$. The learned models are therefore not approximations over spacetime or spiral parameters. They describe one positive-velocity, through-going orbit family on $-17 \leq x \leq 17$, and they are not tested at the open null boundary, at $u_{\text{th}} = 0$, on reflected branches, or on trapped and turning trajectories that may occur in other regimes. Extending the inputs to physical parameters would require new coverage and identifiability studies rather than a simple application of the present checkpoints.

The data are generated from smooth equations with a high-accuracy numerical solver. They contain no observational uncertainty, model discrepancy, or irregular time sampling beyond the deliberately chosen finite-time rows. The reported errors consequently measure approximation of the numerical flow, not robustness to noisy measurements. The inputs also use the exact conserved energy and exact starting states. Behaviour after perturbing $\mathcal{E}_0$, moving an anchor off the exact orbit family, or repeatedly applying the finite-time network is outside the intended use evaluated here.

Neither learned model enforces energy conservation exactly. The transformed coordinate makes admissibility easy to diagnose but does not mathematically prevent an unconstrained linear output from producing $|\xi| \geq 1$; the observed zero violations are empirical. Likewise, the direct model is exact at $s = 0$ by construction but is not constrained to satisfy the semigroup relation

$$\Psi_{s_1+s_2}(z) = \Psi_{s_2}(\Psi_{s_1}(z)).$$

Such a constraint would be relevant if recursive composition of the finite-time network were an intended use. It is not required for the one-shot predictions studied here, and the diagnostic self-composition results do not justify treating the current direct model as a replacement integrator.

The long-time direct-versus-recursive comparison is also conditioned on common survival. At later times, fewer exact anchors remain within their complete orbit and some recursive predictions have already terminated, so the evaluated population changes with s . The frozen crossover definition makes validation and held-out results comparable, but it cannot remove this selection effect. Survival rates and unconditional failure counts should accompany any future claim about recursive performance.

The derivative and split-head experiments remain validation-only. Their main value is explanatory: they show that the rapid-change derivative can be improved and that the lowest-energy residual is predominantly tangential, while the tested late split does not address it. A natural next step would be to model phase along the orbit more directly, or to introduce a loss on short finite-time sequences that measures signed accumulated displacement rather than only pointwise magnitude. Earlier output branching, unequal branch sizes, and separate networks for V_x and F_ξ could then be tested under the same population-level safeguards. These alternatives should be treated as new experiments; the illustrative architectures mentioned above are not results.

Broader physical generalization would require complete-orbit banks over m , ω , α , and θ , including regimes with qualitatively different motion. In that setting, conserved quantities and null-boundary coordinates may still provide useful structure, but their sufficiency would have to be checked again. Uncertainty estimates would also become more useful, especially near sparsely covered boundaries or when an input lies outside the orbit families represented in training.

10. Conclusion

This study began with a short-time neural approximation to a two-dimensional physical flow. One-step accuracy alone did not determine whether repeated prediction would remain physical or preserve the correct trajectory family. A deeper network and the bounded coordinate ξ improved the initial rollouts, while a fixed conserved-energy input produced the largest local-model gain: all 21 representative traversals remained physical and the mean absolute recursive ξ error of the nine difficult incoming cases decreased by 94.5%.

The remaining recursive error was not removed by losses that directly improved local energy-normal or sensitivity-weighted quantities. This led to a direct formulation in which the network predicts the state after an arbitrary elapsed time from an exact anchor. The selected time-rescaled targets enforce identity exactly and balance short- and long-time behaviour more successfully than either raw accumulated increments or pure average rates. On 1,024 sealed held-out orbits, dense RMSE values were 0.06295 in x and 0.002932 in ξ , with no physical violations. The local model remained more accurate at $s = 0.2$; the direct model became more accurate in x after sufficiently long recursive horizons.

The final error analysis shows that the difficult low-energy population remains the limiting case and that much of its scaled phase-space displacement is tangent to the exact orbit. Derivative supervision improves rapid-change and normal errors but leaves this dominant timing component largely unchanged. A late separation of the two network outputs does not improve it over the complete validation population. The main conclusion is therefore not that one network formulation solves every part of the problem, but that physical coordinates, conserved labels, exact endpoint structure, and trajectory-level evaluation each address a different source of error. Keeping these roles separate gives a clearer account of what the models have learned and where their predictions remain limited.

Appendix A. Reduced dynamics and numerical reference

Substituting the rotating spiral into the wormhole line element gives the reduced metric

$$g_{ab} = \begin{pmatrix} -1 + q\omega^2 & q\omega\alpha \\ q\omega\alpha & 1 + q\alpha^2 \end{pmatrix}, \quad q = R(l)^2 \sin^2 \theta,$$

with determinant

$$\det g = -[1 + q(\alpha^2 - \omega^2)].$$

Writing $u = dl/dt$, the radial equation used for the reference trajectories is

$$\begin{aligned} \frac{d^2 l}{dt^2} = & - \frac{\sin^2 \theta (\omega + \alpha u)(b_0^m + l^m)^{2/m-1} l^{m-1}}{1 + \sin^2 \theta (\alpha^2 - \omega^2) R(l)^2} \\ & \times [\alpha u - \omega + 2\omega u^2 + \sin^2 \theta (\omega + \alpha u)^2 \omega R(l)^2]. \end{aligned}$$

Together with $dl/dt = u$, this gives the first-order system integrated numerically. The timelike margin used in physical checks is

$$C(l, u) = 1 - u^2 - R(l)^2 \sin^2 \theta (\omega + \alpha u)^2,$$

so physical states have $C > 0$. The transformed velocity derivative used in Section 5 follows from $u = c(x) + d(x)\xi$:

$$\dot{\xi} = \frac{\dot{u} - u[c'(x) + d'(x)\xi]}{d(x)}.$$

The complete orbit bank was integrated with DOP853 [2] using the tolerances in Table 1. Across all 6,144 training, validation, and held-out parent orbits, the maximum relative energy drift was 7.794×10^{-11} , the minimum timelike margin was 0.0175647, and the total crossing times ranged from 55.8217 to 92.7979. Every retained orbit reached both boundaries and crossed the throat exactly once. Stored dense-output polynomials were used for later state evaluation, so the finite-time transitions and evaluation grids did not require reintegration.

Appendix B. Model and parameter accounting

For a fully connected layer with n_{in} inputs and n_{out} outputs, the affine map contains $n_{\text{out}}n_{\text{in}}$ weights and n_{out} biases. Table B1 gives the resulting counts for the principal models. All listed networks use tanh hidden activations and a linear output.

Table B1. Principal architectures and trainable parameter counts.

Model	Architecture	Parameters
Early shallow local A64	$2 \rightarrow 64 \rightarrow 2$	322
Early deep local A32 $\times$ 32	$2 \rightarrow 32 \rightarrow 32 \rightarrow 2$	1,218
Early augmented local C32 $\times$ 32	$3 \rightarrow 32 \rightarrow 32 \rightarrow 2$	1,250
Transformed local without energy	$2 \rightarrow 32 \rightarrow 32 \rightarrow 2$	1,218
Transformed local with energy	$3 \rightarrow 32 \rightarrow 32 \rightarrow 2$	1,250
Shared finite-time	$4 \rightarrow 64 \rightarrow 64 \rightarrow 2$	4,610
Late-split finite-time	$4 \rightarrow 64 \rightarrow (32, 32) \rightarrow (1, 1)$	4,546

The selected finite-time checkpoint is the seed-202 model at epoch 569, with SHA-256 checksum `a3ae37ead841a1b5a6d6a754052f54a41a54a7fa20e931df0dd987ac871786d4ac8d02e71bd3560e9ebc3228a4a6`. Its selection used only the validation orbit-averaged standardized target MSE. The fixed-energy local comparator used in the direct-versus-recursive study is seed 101 at its native interval $h = 0.2$, with checkpoint checksum `54a41a54a7fa20e931df0dd987ac871786d4ac8d02e71bd3560e9ebc3228a4a6`.

The ordinary finite-time loss gives equal weight to the two standardized outputs,

$$\mathcal{L}_{\text{state}} = \frac{1}{2N} \sum_{k=1}^N \left[\left(\widehat{V}_{x,k} - \widetilde{V}_{x,k} \right)^2 + \left(\widehat{F}_{\xi,k} - \widetilde{F}_{\xi,k} \right)^2 \right].$$

The output factors s and $g(s)$ are not part of this standardized loss; they reconstruct the physical state after the outputs are returned to their original units. The derivative term in Section 5 differentiates through input standardization and through the network, so $\partial_s \widehat{F}_{\xi}$ is the derivative with respect to physical elapsed time rather than the standardized time coordinate.

Appendix C. Finite-time sampling and short-time behaviour

Each parent orbit contributes 64 globally distributed transitions and 32 additional transitions. The 64-row component uses one interior anchor in each of 64 equal x intervals, paired with elapsed-time fractions covering identity, very short, intermediate, long, near-endpoint, and exact-endpoint cases. For $u_{\text{th}} \leq 0.30$, the additional rows place eight anchors on $-17 \leq x \leq -8.5$, with one target in each of four regions: within the compressed incoming branch, immediately after that branch, around the throat, and on the long outgoing branch. For the remaining orbits, the additional rows form a second independent global design. No throat-velocity label is supplied to the network; it is retained only for stratification and evaluation.

Table C1 shows the deterministic short-time comparison that motivated the rate-like targets. The raw accumulated-increment model has a nonzero residual at the identity. Both factored models are exact at $s = 0$, and their errors grow continuously from zero. The pure-rate model is more accurate over most of this very short interval, while Table 5 shows why the time-rescaled ξ target is preferred once intermediate and long times are included.

Table C1. Short-time RMSE for the accumulated, average-rate, and time-rescaled target representations.

	Accumulated		Time-rescaled		Accumulated		Time-rescaled	
s	x	Rate x	x		ξ	Rate ξ	ξ	
0	0.0821655	0	0		0.00258412	0	0	
0.001	0.0821615	8.7917×10^{-7}	1.34357×10^{-6}		0.00258076	3.16544×10^{-7}	4.96329×10^{-7}	
0.01	0.0821262	8.7882×10^{-6}	1.33902×10^{-5}		0.00255096	3.16008×10^{-6}	4.96927×10^{-6}	
0.1	0.0818678	8.97267×10^{-5}	1.29983×10^{-4}		0.00229469	3.11570×10^{-5}	5.04465×10^{-5}	
0.2	0.0817772	1.90523×10^{-4}	2.53557×10^{-4}		0.00209462	6.16326×10^{-5}	1.03067×10^{-4}	
1	0.0870185	0.00145789	0.00119803		0.00185102	0.000274549	0.000599459	

The matched recursive comparison uses the same exact anchors for every method. Table C2 shows how the common population and errors change over a subset of the stored horizons. The finite-time network recursively composed at $s = 0.2$ is omitted because such composition is not its intended use.

Table C2. Validation common-survivor errors for direct prediction and recursive local prediction.

s	Common states	Direct x RMSE	Recursive-local x RMSE	Direct ξ RMSE	Recursive-local ξ RMSE
1	18,555	0.00077412	0.00079151	0.00029966	0.000097963
5	17,527	0.0040867	0.0025813	0.0011463	0.00047574
10	16,496	0.0089074	0.0045178	0.0018757	0.0010376
20	15,240	0.032029	0.026923	0.0047836	0.0080921
30	11,969	0.057654	0.180990	0.0054413	0.0271930
40	8,639	0.083766	0.264230	0.0050892	0.0059685
50	6,054	0.103580	0.361020	0.0050839	0.0130330
60	3,198	0.113640	0.325550	0.0019990	0.0004051

The late decrease of the recursive ξ RMSE at $s = 60$ is one example of why these values must be read together with the common count. It reflects the smaller surviving cohort rather than recovery of the recursive method over the original anchor population.

Appendix D. Scope of the secondary validation experiments

The rapid-change threshold and feature windows were fixed from exact training dynamics before the derivative-loss comparison. Four weights, 0, 0.011, 0.034, and 0.068, were trained for seeds 101, 202, and 303. The selected value 0.034 was chosen from validation by the joint improvement of the rapid-feature derivative and post-feature state errors, subject to global x and physical safeguards. The larger weight was rejected because of its global x degradation. Feature-level associations between derivative improvement and subsequent state improvement were weak, so the aggregate change is not interpreted as a causal one-feature relation.

The phase-space tangent/normal study uses errors and exact tangents scaled by the training standard deviations in Section 5. Its lowest-bin post-change comparison includes all available validation orbits and seeds, rather than only the named $u_{\text{th}} = 0.05$ reference. The continuous curve in Figure 8 ends at 0.15 because no post-change rows are present in $0.15 < u_{\text{th}} \leq 0.30$. Values beyond that gap were not interpolated.

The split-head comparison trained both $\lambda_{\xi} = 0$ and 0.034 versions for all three seeds and reused the corresponding shared checkpoints. The conclusion in Section 7 is based on the latter pair because it asks whether output specialization improves the remaining residual after the selected derivative treatment. No $\lambda_{\xi} = 0.068$ split model and no alternative branch allocation was trained. None of the derivative or split-head models was evaluated on the sealed held-out cohort.

References

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